

CS173 Course Project

Requirements

- Using data provided and/or data collected by yourselves
- A research project or application project
 - data analysis, visualization, and modeling
 - examining real-world datasets and associated problems.
 - Research: new problem or/and new method
 - Application: solving existing problem, with solid adaptation of methods improved or tailored to this problem. Thoroughly compare the methods. (data challenge)

Requirements

- 4 people in a team (5-7 teams): **proposal**, **interim**, and **final** reports/presentations
 - Proposal: report, presentation, rate each other's project
 - Interim: report, presentation, rate each other's project
 - Final: **code**, report, presentation, rate each other's project
- **3/16 Mon** before 23:59 pm, submit team information: member names, project name
- **3/29 Sun** before 23:59 pm, submit **proposal** report on Blackboard
- **3/30 Mon** (5th week) **proposal** presentations
- 4/22 Wed before 23:59 pm, submit **interim** report on Blackboard
- 4/23 Thu (8th week) **interim** presentations
- 5/20 Wed before 23:59 pm, submit **final** report/slides/code on Blackboard
- 5/21 Thu (12th week) **final** presentations

Bonus

- $\leq 2.5\%$ of the total score for the course for MapReduce
 - For the projects that include **MapReduce** for data processing or modeling in the **final** reports/presentations
 - Submit the **MapReduce** code in the final submission
 - In the **final** report, create a separate section to describe the scenario in which MapReduce is used, its logic, its running version and environment, and compare the performance enhancement with regular method without MapReduce.
- 1.5% of the total score for the submission to a [CCF A/B/C conference or journal](#)
 - Submit the **submission receipt email** to us.

Proposal

- Format
 - Chinese or English (Express yourselves clearly)
 - Double columns, 2-3 pages (at least 2 pages) + references
 - **Please use the ACM template, preferably in latex:**
<https://www.acm.org/publications/proceedings-template>
- Content
 - **Title**
 - **Abstract**
 - **Introduction**
 - **Related work**
 - **Problem statement**
 - **Experiment methodology (solution)**
 - Data mining techniques to be used
 - **Feasibility** (data/method, etc.)
 - Evaluation plan
 - **Timetable and workload division**
 - **References**

<https://www.lakeheadu.ca/sites/default/files/uploads/56/GraduateMScResearchProposalTemplate.pdf>

ACM International Conference on Information and Knowledge Management (CCF B)

Last Year's Call for Short Paper

Call for short papers 2025 <https://cikm2025.org/calls/short-research-papers>

4 pages, plus unlimited references.

Abstract May 30, paper June 6

This Year: <https://cikm2026.org/> (call for paper not available yet)

The International AAAI Conference on Web and Social Media (ICWSM) (CCF B)

2027 (next year) first round of submission Mar 11, 2026

2026: <https://www.icwsm.org/2026/index.html>

2026's call for paper:

<https://www.icwsm.org/2026/submit.html#full-papers>

IEEE Big Data 2026 (CCF C)

Call for papers 2026

<https://bigdataieee.org/BigData2026/calls/papers/>

up to 10 page IEEE 2-column format, reference pages counted in the 10 pages

Aug 21, 2026

2025 Spring student paper

Accepted by IEEE BigData Workshop

Dear Guo Siqi and Tang Zhihao,

Congratulations! We are pleased to inform you that your paper has been accepted as a regular paper to be presented in IEEE BigData 2025 Special Session on Large Language Models in Social Data Science.

Paper ID: BigD656

Paper Title: Multi-Layer Feature Fusion for Detecting AI-Generated Code in Programming Education

Authors: Guo Siqi and Tang Zhihao

IEEE BigData 2025 will be held in Macau SAR, China, 8–11 December, 2025.

Please revise your paper carefully to address the reviewers' comments and suggestions.

Please carefully read the important information about CAMERA-READY PAPER and COPYRIGHT SUBMISSION, PAPER REGISTRATION, and PAPER PRESENTATION on the homepage (<https://conferences.cis.um.edu.mo/ieeebigdata2025/index.html>)

If you have any question, please feel free to contact us via: qywang@ctbu.edu.cn or zwguo@ctbu.edu.cn.

We are looking forward to seeing you in Macau SAR, China, 8–11 December, 2025.

Best regards!

Program Chair

Abstract

- Be one paragraph long
- Provide a high-level summary of your project and outline your main goals.

Introduction

- Introduces the project and what you are trying to do. Should motivate the problem, quickly define it and the approach taken, and may discuss some highly related work. Probably should mention and contributions of the work.

Related work

- A description of related work, with citations to relevant papers. If you are doing an application paper, where you analyze some data, you are less likely to rely on a lot of related work. Nonetheless, there almost always should be some related work discussed.
- You may want to provide the bulk of related work in this section, but include some in the introduction too. There almost always will be related work, even if it is not a perfect match. All related work papers should be cited in the paper and then should appear in the reference section. All items in the reference section should be explicitly cited in the paper.

Problem statement

- Formally describe the problem you are going to solve, **preferably with mathematical definitions** (you may see how other papers define their problems).

Experiment methodology

- Describes the experiments and the experiment methodology. Will describe the **datasets**, **evaluation metrics**, data mining algorithms to be used, visualizations to be created, the **methodology related to the setup of experiments**, and any other details related to the experiments. There will usually be a subsection for each of the sub-topics just mentioned.
- For proposals, we are not expecting very detailed descriptions. But we need to know your plan. The plan should be feasible, which means the **data is feasible** and the **idea and methods are sound**.

Timetable and workload division

- The timetable should be fine-grained, preferably with a milestone for each two weeks.
- Should describe how you divide the work among all the team members. The division should be reasonable.

Evaluation

- 项目总分占课程总分的45%+4%（项目42%+互评3%+bonus 4%）
- 开题、中期、结题分别占项目的20%,30%与50%，其中报告占70%，presentation 占30%。
- 开题报告的评分方式：
 - 形式：规范性 20%
 - 内容：翔实，准确，逻辑清晰， 25%
 - 方法：理论基础完善，方法合理，包含数据挖掘方法 25%
 - 可行性： 15%
 - 创新性： 15%
- 开题presentation内容应涵盖开题报告内容
- presentation的评分方式：
 - PPT质量（整体结构清晰，内容完整、简明，图表、数字准确传达） 50%
 - 语言表达（条理性、说服力、问答环节表现） 50%
 - Tips <https://www.icetl.org/wp-content/uploads/2021/04/6-Tips-for-Giving-a-Fabulous-Academic-Presentation-ACAVENT.pdf>
 - 6分钟presentation, 4分钟QA

Interim report

- Content
 - Please include necessary contents from the proposal
 - Progress (What you did, referring to your previous timetable)
 - Outcomes (Results)
 - Issues and Challenges
 - (Adjusted) plan and timetable

Final report

- In the format of an academic paper
 - Refer to: Zobel - Writing for computer science 3rd edition
 - Some samples
 - [Mapping the world's photos](#)
 - [Inferring social ties from geographic coincidences](#)

CS173 previous projects

2022 projects

- Prediction of the number of new cases of COVID-19 based on machine learning
- Analyze and Predict NBA players' On-court Performance and their Impact on Teams
- Visualize and Predict Terrorist Attacks using the Global Terrorism Database
- Restaurant Recommendation System by Analysing Restaurant Attributes and Individual Preference
- A Deep Learning Pipeline Framework for the Prediction of Stock Prices
- Citation-based Paper Ranking with AutoEncoder
- Traffic Prediction based on Spatio-Temporal Modeling and Alternative Data Mining

2023 projects

- Linking Tweets and News Articles using BERT and ChatGPT
- Monitoring Mental Health through Social Media
- 基于 **CBDB** 的中国古代官员社会关系与地理分布研究： 以宋代为例
- LightFM-SVD:Hybrid E-commerce Recommendation System
- Does the ideological bias of the US mass media dominates their attitudes towards China?

2024 projects

- Walking says something about your state of mind: A study of student behavior based on the StudentLife Dataset
- Quantification and Prediction of Terrorist Attack Risk Under Geographic and Temporal Dimensions
- Which Comment should I Look? A Data Driven Analysis on Reviews from the Developers' Standpoint
- What a witty comment! Identify Clever Comments in Online Media Platforms
- Exploring the Factors Affecting the Appointment and Promotion of Officials in the Sui and Tang Dynasties Using CBDB

2025 projects

- Detecting AI-Generated Short Algorithmic Code
- PathSeek: an LLM-Enhanced path finding method for new hypothesis generation in biomedical knowledge graph
- Multimodal Fake News Detection via LVLM-based Feature Learning and Modal Fusion
- The impact of demographic, economic, and social data on U.S. election results by state
- 基于微博博文情绪分析的话题热度趋势预测研究

CS277 previous projects

2019 projects

- Forecasting Agricultural Commodity Price with Internet Search Activity
 - Baidu index data, online financial forum discussions, commodity price data
- Predicting Movie Box Office based on Social Media Information
 - Movie ratings, movie reviews, etc., from Douban
- Deep Reinforcement Learning in Stock Portfolio Management
 - SSE stock information
- Predicting Minute-Level Stock Prices Based on LSTM
 - US stock information

2021 Spring projects

- Knowledge-based Stock Trend Prediction
 - Use KG to represent relationships between listed companies
- Deep Reinforcement Learning in Stock Portfolio Management
- Predicting Minute-Level Stock Prices Based on LSTM

What did we do?

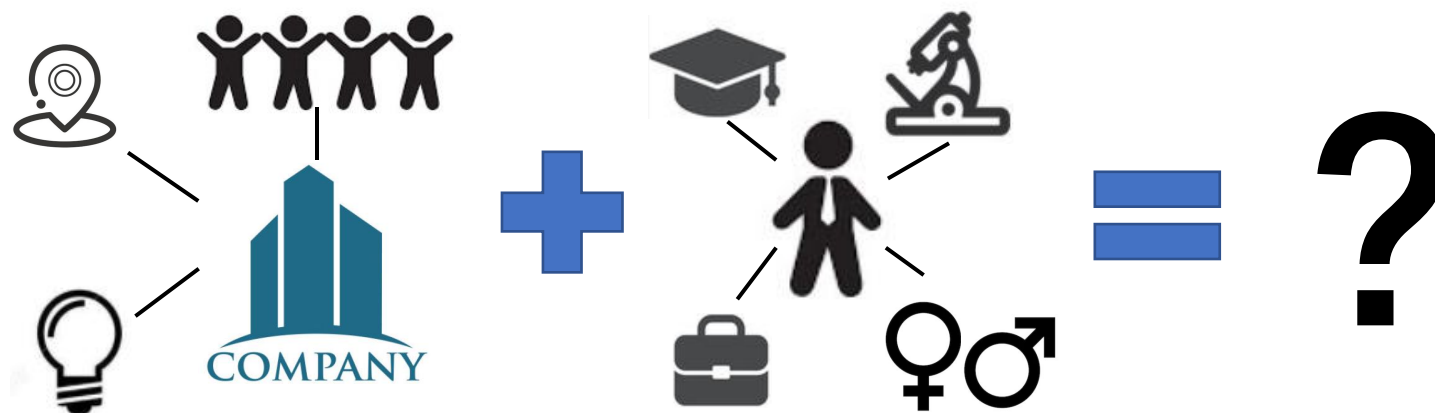
初创公司VC投资成功预测

我们做了什么？

预测：
一笔投资五年内能否
成功退出

我们做了什么？

20万笔投资, 12万创业公司, 7万投资人, 27万创业者



建立机器学习/深度学习模型，挑选优质创业公司

模型表现

- 基于历史数据建模，从2014年7037家获首轮融资的公司中选择高质量公司，观察其5年内的成功率（IPO或被收购）
- 投资人实际表现：

7037家 $\xrightarrow{\text{投资人选中进行第二轮投资}}$ 5219家 $\xrightarrow{\text{5年内成功}}$ **11%** (574/5219)

- 我们模型成功率是投资人的**3倍以上**：

7037家 $\xrightarrow{\text{模型选择}}$ **120家** $\xrightarrow{\text{5年内成功}}$ **36%** (43/120)

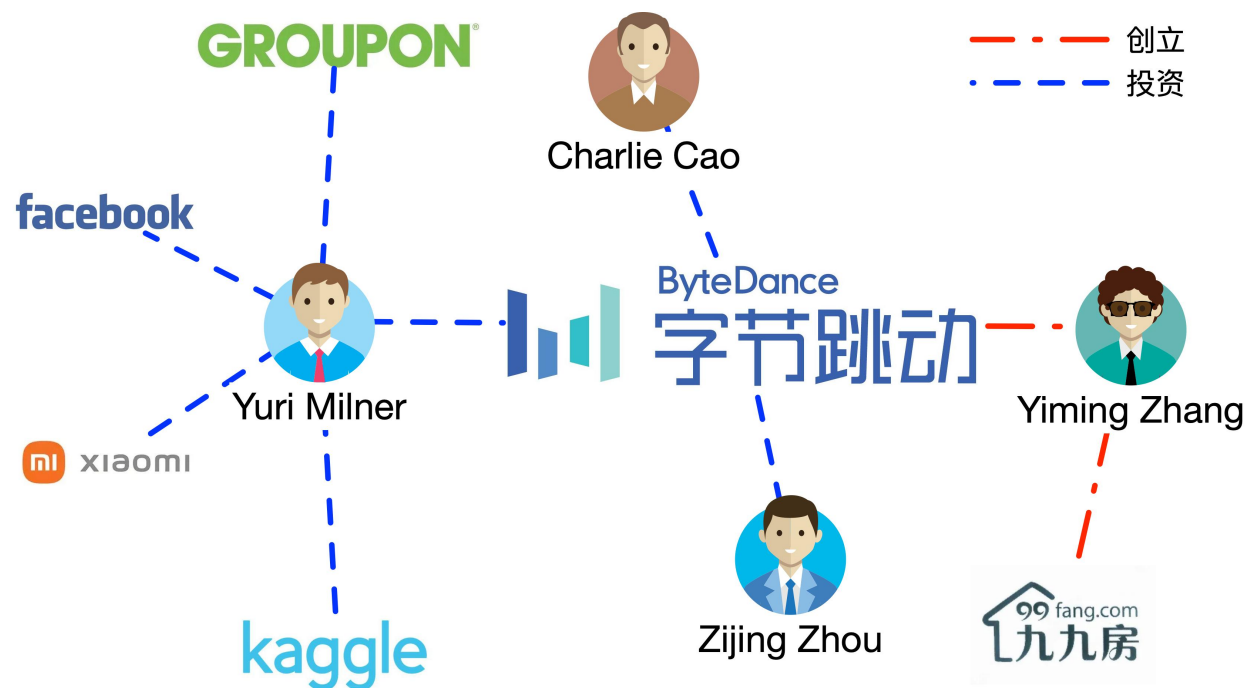
7037家 $\xrightarrow{\text{模型选择}}$ **600家** $\xrightarrow{\text{5年内成功}}$ **34%** (204/600)

动态图神经网络（GST）——结构与时序

- 预测：首轮融资时点，5年内成功（并购/IPO）
- 数据源：PitchBook（晨星旗下），全球VC投资，1977—2019
- 覆盖：116,764家初创公司；187,346笔投资；244,267位创始人/联合创始人；62,424位VC投资人
- 粒度：交易级细节（投资方、金额、轮次、日期）；人/公司画像（地域、教育、履历）
- 实体与字段：
 - 公司：ID、名称、成立年、城市/国家、行业（三级）
 - 个人：ID、姓名、学位、院校、毕业年、性别
 - 投资：Deal ID、类型/轮次、投资方、日期、金额、Company ID

[Help Me Screen: Analyzing and Predicting the Success of Start-ups in Dynamic Venture Capital Networks](#)

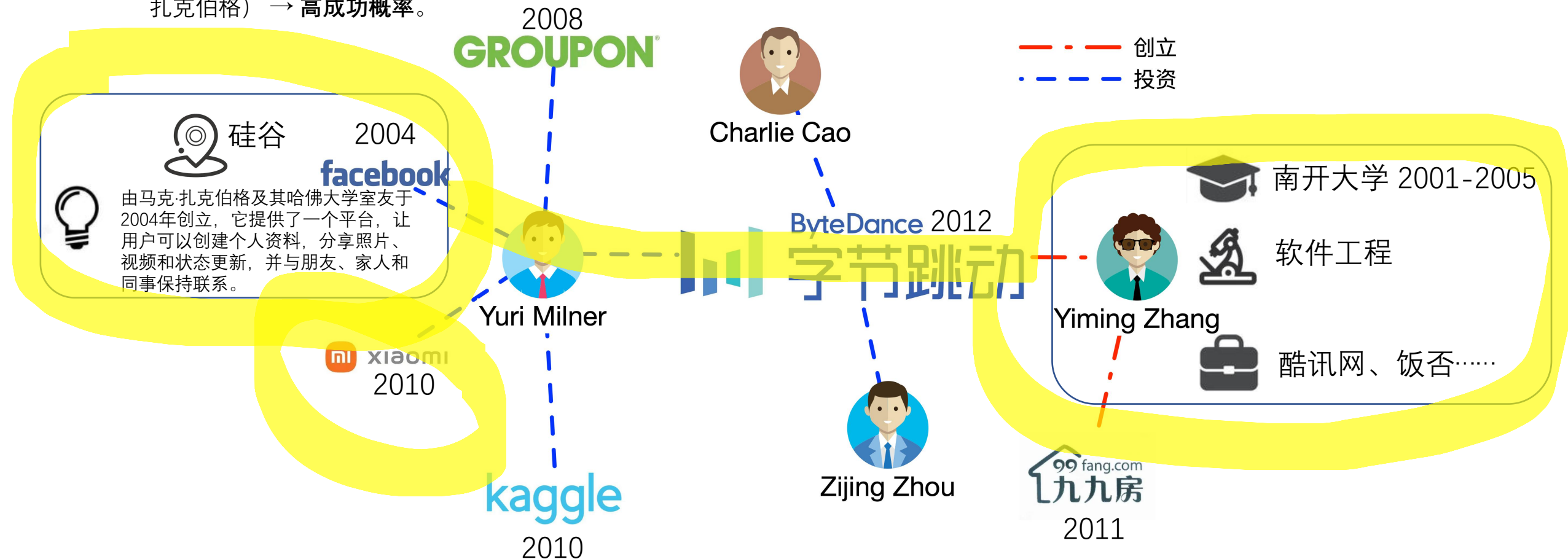
- 网络结构



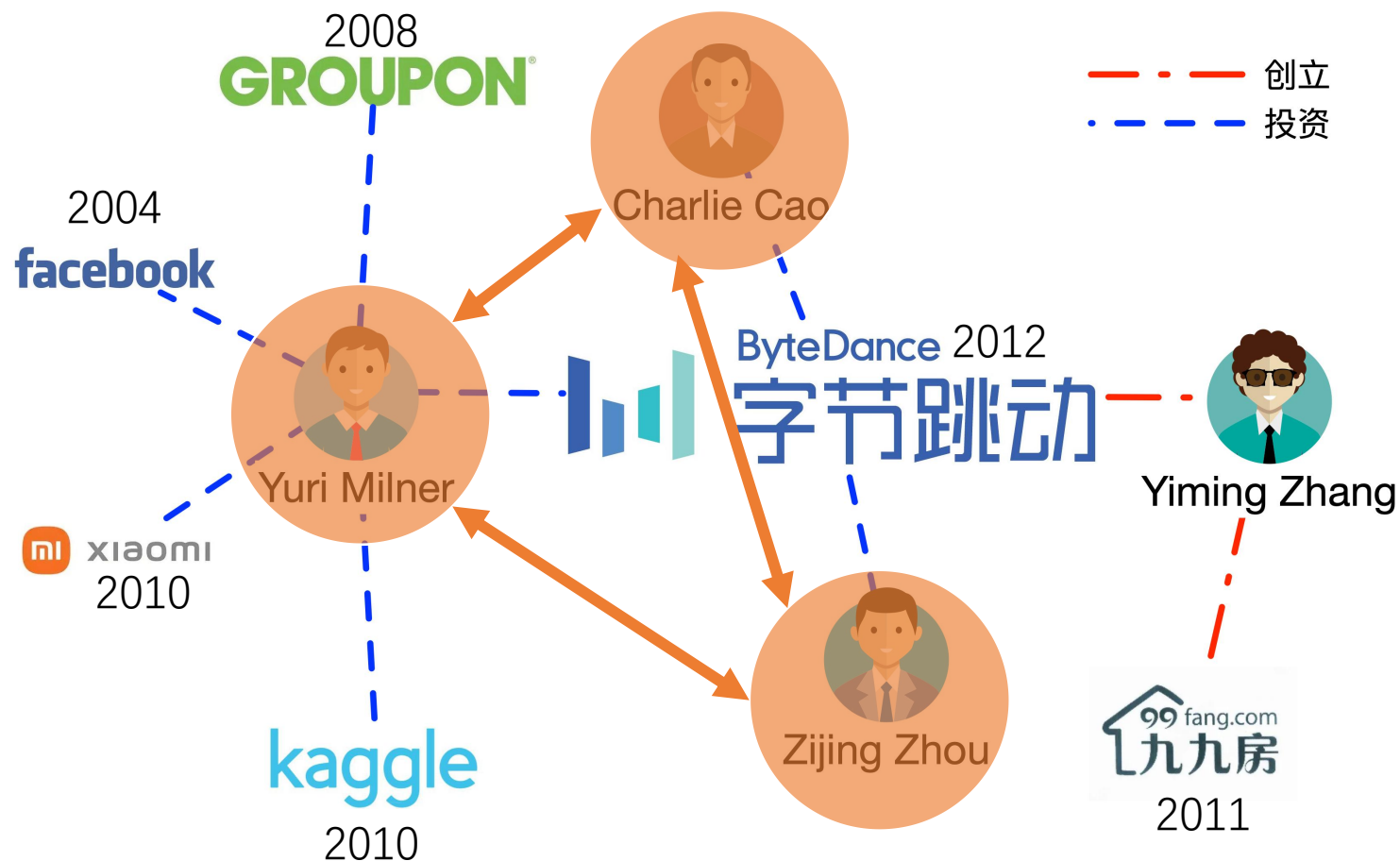
- 节点：
 - 人员（投资人、创业者、管理层人员、董事会成员、顾问）
 - 公司（创业公司、创业公司成员任职过的公司、投资机构）
- 边：
 - 雇佣与投资关系
- 按月更新网络

• 融合文本语义的多跳推理

- 字节能否成功？Yuri Milner成功先验（Facebook、小米）→ 高声誉合投 → 中国公司共性（字节×小米：工程驱动、快速迭代、强供应链/分发能力、本土高增长）→ 字节×Facebook共性（算法信息流、社交内容平台、广告变现）→ 创始人强产品工程（张一鸣≈扎克伯格）→ 高成功概率。



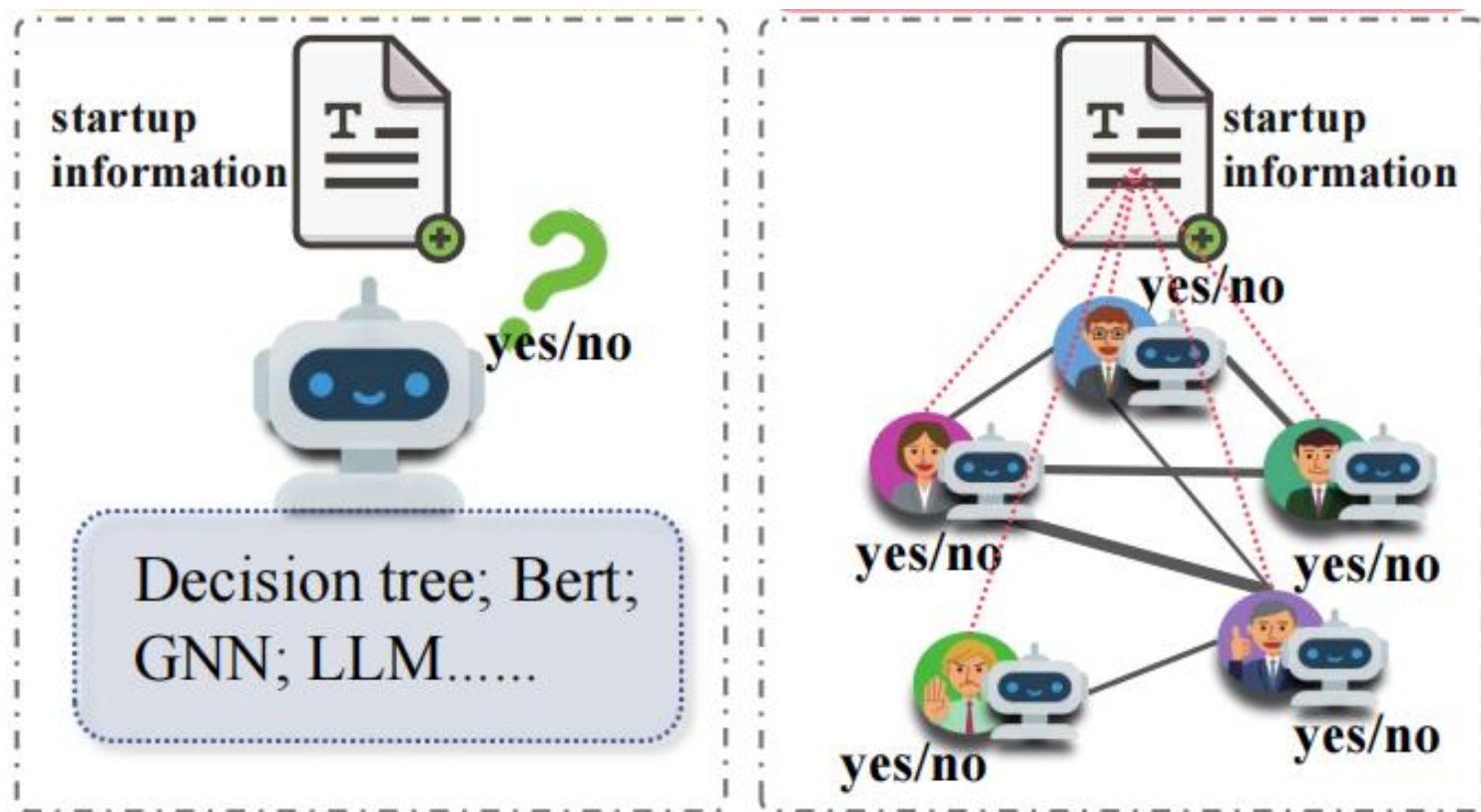
- 投资人信息交互
- VC决策常由多名潜在投资人相互影响形成合投与共识 (Han & Yang (2013), Liu (2017))



[LLM Agents as VC investors: Predicting Startup Success via RolePlay-Based Collective Simulation](https://arxiv.org/pdf/2512.22608)

<https://arxiv.org/pdf/2512.22608>

传统单决策 vs 群体交互决策

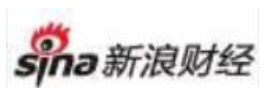


合投者来自紧密群体：合投投资者平均图距离=2.16；全网平均=4.23

基于知识图谱的银行间市场智能监测与风控平台研发与应用

中国外汇交易中心中汇公司

构建外汇市场参与者知识图谱，量化
分析舆情对相关机构的影响面与影响
力，防控系统性金融风险



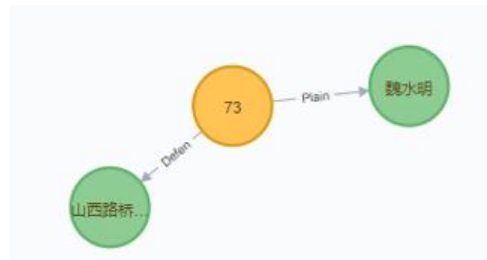
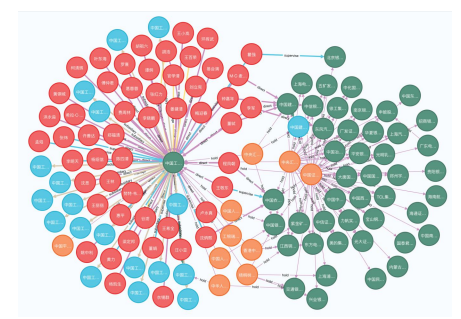
- 1. 异构舆情数据源
爬取与清洗
- 2. 基础图谱构建

标题：包商银行两员工受贿50余万，违法发放2亿
贷款，多年未能收回
...裁判文书网发布了河北省衡水市中级人民法院对刘京鹏、牛敏非国家工作人员受贿、违法发放
贷款的二审刑事判决书。而包商银行上一次受到广
泛关注，是迟迟未能发布年报。《每日经济新闻》
记者就年报信息迟迟未披露等问题采访包商银行。
但在截稿前，该行并未做出任何回复。...违法发放
贷款2亿收了53.2万“咨询费”...

2. 舆情多维解读

3. 公告风险事件抽取，
形成新关系，纳入基础
图谱

4. 多维数据完成
知识图谱构建



1. 外汇市场金融知识图谱建设与应用 I



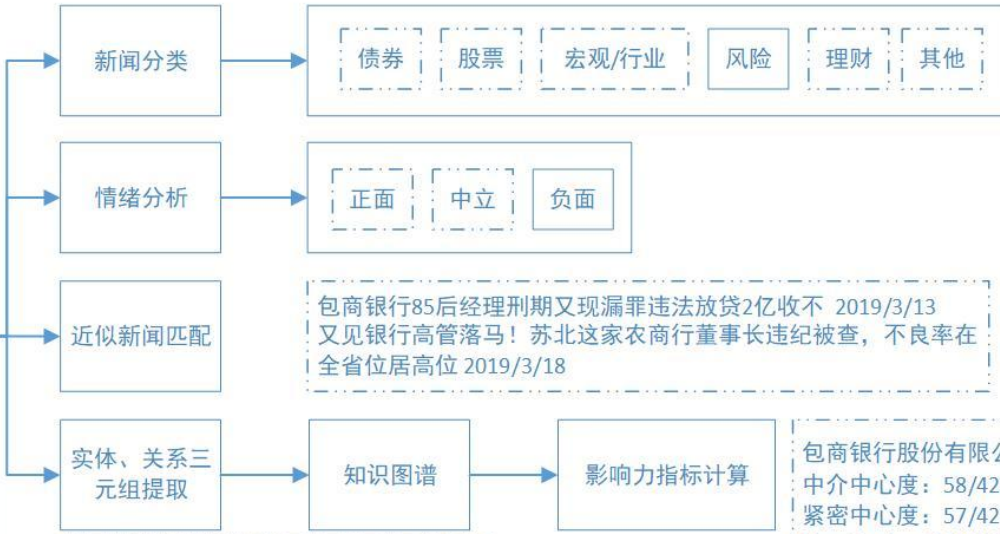
实时捕捉会员风险舆情，估计其对系统造成的影响

- 1. 对舆情进行分类、情感分析、近似新闻匹配、命名实体识别
- 2. 舆情关联至会员股权图谱，中心性指标量化分析影响面与影响力

多源头新闻获取

标题：包商银行两员工受贿50余万，违法发放2亿贷款，多年未能收回
...裁判文书网发布了河北省衡水市中级人民法院对刘京鹏、牛敏非国家工作人员受贿、违法发放贷款的二审刑事判决书。而包商银行上一次受到广泛关注，是迟迟未能发布年报。《每日经济新闻》记者就年报信息迟迟未披露等问题采访包商银行。但在截稿前，该行并未做出任何回复。...违法发放贷款2亿收了53.2万“咨询费”...

新闻

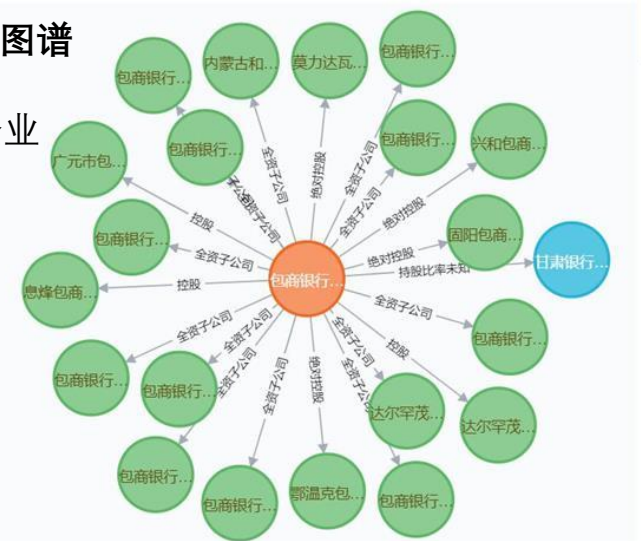


舆情解读

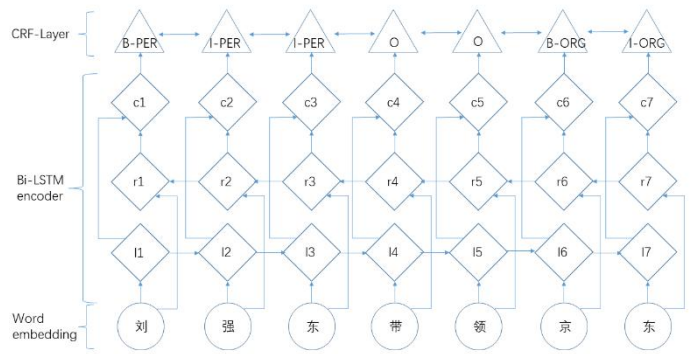
模块	方法
新闻分类	BoW + SVM
情感分析	LSTM
相似新闻	TF-IDF+CosSim
命名实体识别	BiLSTM-CRF

构建会员股权图谱

843家会员
3795家关联企业
3939人
9831条关系



文本涉及实体以及关联公司可视化



Bi-LSTM-CRF进行命名实体识别

中心度量化分析机构影响面与影响力

中介中心度

$$C_B(v) = \sum_{s \neq v \neq t \in V} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

紧密中心度

$$CC(i) = \frac{1}{d_i} = \frac{n-1}{\sum_{j \neq i} d_{ij}}$$

PageRank中心度

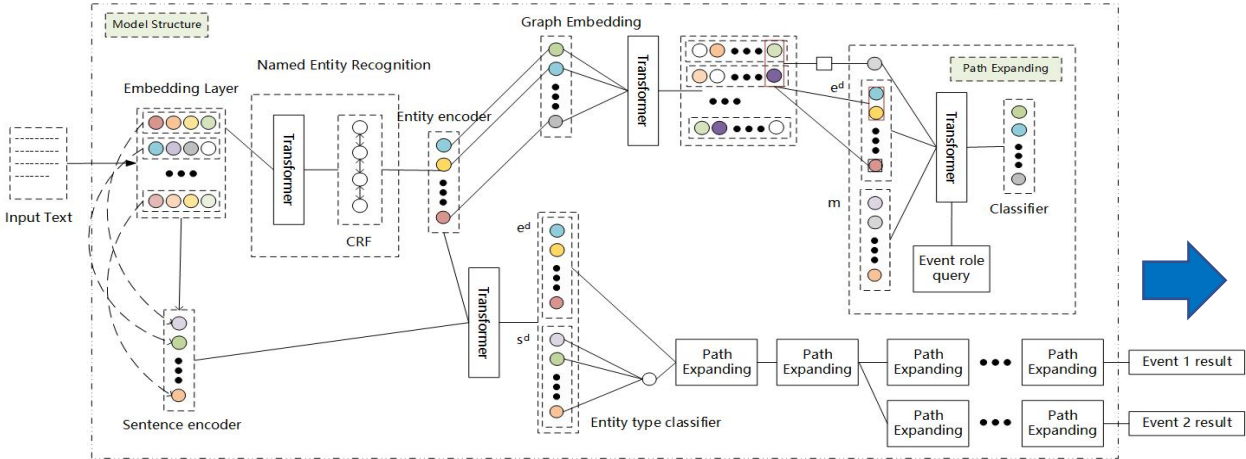
$$P_{n+1} = AP_n$$

对金融文本（上市公司公告）的篇章级多事件抽取，发现风险信息，补充外汇市场参与者图谱
20%会员机构为上市公司；绝大部分会员与上市公司有借贷等业务往来

采集2010年以来**338万**余条上市公司公告

基于知识增强的事件抽取模型

601566:九牧王关于股东股份质押的公告
“九牧王股份有限公司（以下简称“公司”）股东**泉州市铂锐投资管理有限公司**（以下简称“铂锐投资”）持有公司股份 25,150,000 股，占公司总股本的 4.38%，本次**质押股份 21,186,442 股**，占其持有公司股份数的**84.24%**。公司股东铂锐投资及其一致行动人**陈彬彬**合计持有公司股份30,150,000 股，占公司总股本的 5.25%。铂锐投资及其一致行动人累计**质押股份 21,186,442 股**，占其持有公司股份数的 **70.27%**。...股质押给**中国国际金融股份有限公司**”
标注股份质押事件：
股东*2，质押方，质押股数，质押比例（对应各股东）



- 事件类型
- 股权质押
 - 股权冻结
 - 股份回购
 - 诉讼
 - 高管增持
 - 高管减持

- 引入知识图谱，进行多事件文本抽取
1. 利用Transformer+CRF进行实体识别
 2. 六类风险事件抽取
 3. 工商信息图谱作为先验信息加入模型辅助抽取
 4. **F1值比最优方法提升6.1%，达到81.7%**
- (论文收录于IEEE Big Data 2020)

事件角色	事件元素
出质人	泉州市铂锐投资管理有限公司
质权人	中国国际金融股份有限公司
质押数量	21,186,442股
累计质数	21,186,4442股
质押比例	84.24%
...	

国际政要推文解读与市场预测

Understanding Tweets from Political Figures and Estimating the Markets

- 政要的社交媒体发言，对研判国际局势有重要意义
 - 特朗普“推特治国”
 - 发言可能对金融市场产生影响
- 研究问题
 - 能否智能化解读全球政要推文？
 - 知识库链接、前序新闻挖掘、事理归因、主题标签
 - 能否基于推文理解实现智能问答？
 - 知识图谱、QA系统

研究框架



Donald J. Trump
@realDonaldTrump

I am pleased to report that the U.S. has made substantial progress in our trade talks with China on important structural issues including intellectual property protection, technology transfer, agriculture, services, currency, and many other issues. As a result of these very.....

2:39 PM - 24 Feb 2019

....productive talks, I will be delaying the U.S. increase in tariffs now scheduled for March 1. Assuming both sides make additional progress, we will be planning a Summit for President Xi and myself, at Mar-a-Lago, to conclude an agreement. A very good weekend for U.S. & China!

2:50 PM - 24 Feb 2019

21,275 Retweets 104,171 Likes

12K 21K 104K

命名实体及关系识别

推文情感分析

联系实时行情

I am pleased to report that the U.S. has made substantial progress in our trade talks with China on important structural issues including intellectual property protection, technology transfer, agriculture, services, currency, and many other issues. As a result of these very productive talks I will be delaying the U.S. increase in tariffs now scheduled for March 1. Assuming both sides make additional progress we will be planning a Summit for President Xi and myself at Mar-a-Lago to conclude an agreement. A very good weekend for U.S. and China!

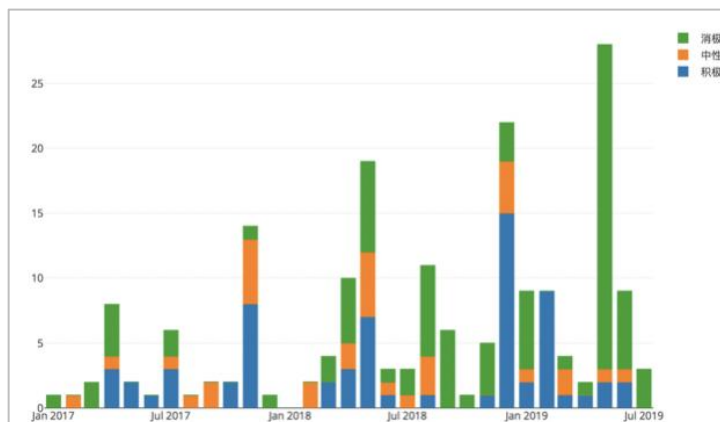
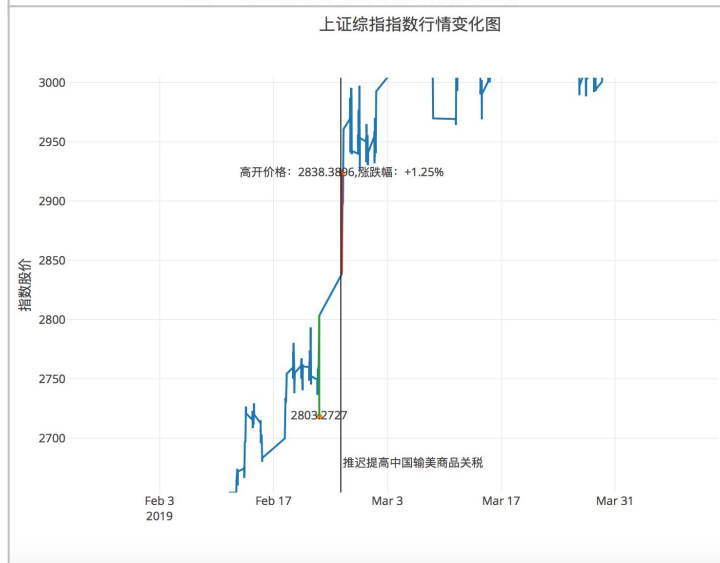


图 1 特朗普与中国相关推文-情绪堆积图



实体链接
外部知识进行
数据补充

新闻/事件匹配
实时新闻帮助
理解推文

About: Intellectual property

An Entity of Type : Thing, from Named Graph : <http://dbpedia.org>, within Data Space : dbpedia.org

Intellectual property (IP) refers to creations of the intellect for which a monopoly is assigned to designated owners by law. Intellectual property rights (IPRs) are the protections granted to the creators of IP, and include trademarks, copyright, patents, industrial design rights, and in some jurisdictions trade secrets. Artistic works including music and literature, as well as discoveries, inventions, words, phrases, symbols, and designs can all be protected as intellectual property.

Statement on Steps to Protect Domestic Technology and Intellectual Property from China's Discriminatory and Burdensome Trade Practices

ECONOMY & JOBS | Issued on: May 29, 2018

On March 22, 2018, the President signed a memorandum announcing that the United States would take multiple steps to protect domestic technology and intellectual property from certain discriminatory and burdensome trade practices by China. These actions were announced following a report of the Office of the U.S. Trade Representative regarding China's practices with respect to technology transfer, intellectual property, and innovation. In accordance with the March 22 memorandum, the President has been updated on the progress of the announced actions as follows:

推文解读：推理推文内容的因果关系，实现图谱对非结构化文本的表示
股市预警：对股票市场波动进行预警

Provided data
(course-usage only)

- **Cultural and art data**
 - IMDB movies
 - Chinese poems and poets
 - NBA data
 - Academic paper data
 - 中华家谱总目
- **Fundamental data**
 - Wolfram Alpha
 - Wikipedia
 - GTD (Global Terrorism Database) and GDELT (Global Database of Events, Language, and Tone)
 - Geonames
- **News and social media**
 - Chinese financial news
 - Reuters, New York Times, Jpost news
 - Twitter
- **Financial Data**
 - VC data (Crunchbase)
 - Listed company fundamental data (新浪财经)
 - Analysts' reports (券商分析师研究报告)
 - Listed company announcements
 - Market Data
 - 天眼查数据
- **Socio-economic data**
 - Nighttime light intensity
 - Other socioeconomics data

Cultural and Art Data

中国历代人物传记资料库 (CBDB)

- 哈佛大学从中国的古代人物传记中整理出515,488人的信息，构建了[中国历代人物传记资料库 \(CBDB\)](#)。
- 时间跨度约为7世界至19世纪，收集了史料中记载的历史人物的基本信息、社会关系和入仕情况等数据，使用该数据集可以用于群体传记学的统计、网络 and 空间分析。
- “由人物 (People)、親屬 (Kinship)、非親屬社會關係 (Non-kinship Associations)、社會區分 (Status)、入仕途徑 (Modes of Entry into Government)、宦歷 (Offices / Postings)、地址 (Places)、著述 (Writings) 等部份組成。”
 - [数据下载](#)
 - [数据详细描述](#)
 - [数据表链接关系](#)

APS (American Physical Society) Data

- Data available in: [Course Server](#)
 - There are many classified folders in the metadata folder, which can help us index specific paper information (in each json file).
 - e.g., aps-dataset-metadata 2020/PR/1/PhysRev.1.1.json
 - PACS.txt describes the code and paper corresponding classifications.
 - The citation relations are recorded in aps-dataset-citations-2020.csv file.

Description	Metadata
Records	678,916
Time Range	1893~2020
Content (id/DOI code is the the unique identifier)	id, title, publisher, journal, issue, volume, pageStart, pageEnd, seqnum, date, numPages, articleType, identifiers, rights, authors, affiliations

Description	Metadata
Records	441,341
Content (Each paper has at most 5 codes)	DOI, PACS1, PACS2, PACS3, PACS4, PACS5

Description	Citations
Records	8,850,335
Content	citing_doi, cited_doi
Citation example:	
citing_doi	cited_doi
10.1103/PhysRevSeriesI.11.215	10.1103/PhysRevSeriesI.1.1

NBA Data

- Dataset with all NBA games from 2004 season to dec 2020 ([Source url](#) or [Course Server](#))
 - games.csv : all games from 2004 season to last update with the date, teams and some details like number of points, etc.
 - games_details.csv : details of games dataset, all statistics of players for a given game
 - players.csv : players details (name)
 - ranking.csv : ranking of NBA given a day (split into west and east on CONFERENCE column)
 - teams.csv : all teams of NBA

IMDb Dataset

- IMDb is the world's most popular and authoritative source for movie, TV and celebrity content.
- Data description and download url can be found in ([Kaggle](#))
- Data available in ([Course Server](#))

Subset name	title.akas	title.basics	title.principals	name.basics	title.ratings
Records	19,322,610	6,321,302	36,468,817	9,706,922	993,153
Description	The same film may have multiple titles, this file contains alternate title information.	This file describes basic information about the video titles.	This file contains principal cast/crew members such as actors, writer etc. which are associated with each film.	This file contains professions' birth and death year info as well as their most known work of art.	This file contains movie rating entries.

搜韵

- 搜韵约84万诗词，可从作者，朝代，韵部，关键词位置等多个角度进行模糊或准确搜索。[文档链接](#)
- 数据通过访问API获取。
 - `https://api.sou-yun.cn/open/poem?key={} &dynasty={} &scope={} &type={} &json_type={} &pageno={}`
 - key, dynasty, scope, type, json_type, pageno是输入的搜索参数.
 - 推荐使用[Regeust](#) and [XML](#) python 库进行数据收集和处理.

名字	历代诗词	相关词条用韵	古人诗例用韵	对仗词汇
数据量	800,000	400,000	1,200,000	1,800,000

This XML file does not appear to have any style information associated with it. The document tree is s

```
<?xml version="1.0" encoding="UTF-8" ?>
<QueryPoemResult xmlns:i="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://schemas.datacontract.org/2004/07/PoemApplication.Comm
  <Authors i:nil="true"/>
  <Count>722</Count>
  <Keys xmlns:d2p1="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
    <d2p1:string>楼</d2p1:string>
  </Keys>
  <Notification i:nil="true"/>
  <PageNo>1</PageNo>
  <PageSize>20</PageSize>
  <Poems i:nil="true"/>
  <ShiData xmlns:d2p1="http://schemas.datacontract.org/2004/07/PoemApplication.Comm
    <d2p1:Shi>
      <d2p1:Allusions xmlns:d4p1="http://schemas.datacontract.org/2004/07/PoemApplication.Comm
        <d2p1:Author>沈佺期</d2p1:Author>
        <d2p1:AuthorDate i:nil="true"/>
        <d2p1:AuthorId>17898</d2p1:AuthorId>
        <d2p1:AuthorPlace i:nil="true"/>
      </d2p1:Shi>
    </d2p1:Shi>
  </ShiData>
  <Clauses>
    <d2p1:Ju>
      <d2p1:BreakAfter i:nil="true"/>
      <d2p1:Comments i:nil="true"/>
      <d2p1:Content>主家天上凤楼人，</d2p1:Content>
      <d2p1:Tones>0</d2p1:Tones>
    </d2p1:Ju>
    <d2p1:Ju>
      <d2p1:BreakAfter i:nil="true"/>
      <d2p1:Comments i:nil="true"/>
      <d2p1:Content>传得仙音到客新，</d2p1:Content>
      <d2p1:Tones>0</d2p1:Tones>
    </d2p1:Ju>
    <d2p1:Ju>
      <d2p1:BreakAfter i:nil="true"/>
      <d2p1:Comments i:nil="true"/>
      <d2p1:Content>未省落梅和折柳，</d2p1:Content>
      <d2p1:Tones>0</d2p1:Tones>
    </d2p1:Ju>
    <d2p1:Ju>
      <d2p1:BreakAfter i:nil="true"/>
      <d2p1:Comments i:nil="true"/>
      <d2p1:Content>风吹散作万家春，</d2p1:Content>
      <d2p1:Tones>0</d2p1:Tones>
    </d2p1:Ju>
  </d2p1:Clauses>
  <d2p1:Comments xmlns:d4p1="http://schemas.datacontract.org/2004/07/PoemApplication.Comm
    <d2p1:Dynasty>初唐</d2p1:Dynasty>
    <d2p1:Froms xmlns:d4p1="http://schemas.microsoft.com/2003/10/Serialization/Arr
      <d2p1:GroupIndex>0</d2p1:GroupIndex>
      <d2p1:Id>724325</d2p1:Id>
      <d2p1:IsTwoClausesPerSentence>true</d2p1:IsTwoClausesPerSentence>
      <d2p1:Links xmlns:d4p1="http://schemas.datacontract.org/2004/07/PoemApplication.Comm
```

中华家谱总目

- 《中国家谱总目》数据集共1200万字，共收录海内外收藏的中国家谱条目共52401条，姓氏计608个。《中国家谱总目》是迄今为止收录中国家谱数量最多、姓氏最多、参与编纂机构最多的目录，在中国谱牒目录学上具有划时代的意义。
- 由居地，书名，堂号，版本年代，迁徙信息，收藏地点，始祖，始祖年代等部分组成。
- 使用该数据集可用于古代人口迁徙，姓氏时空分布等问题的研究分析。
- 数据示例：

居地	书名	版本年代	责任者	责任者时代	始祖	始祖時代	迁徙信息、家族源流	
江蘇淮安	山陽丁氏族譜不分卷	清咸豐三年	丁晏	清	國信	明	始祖國信，明代人。始遷祖近淮，明代人。清人丁晏、丁壽昌即出於是族。	

- [数据详情](#)
- [下载地址](#)

Fundamental Data

Wolfram Alpha

- Computational knowledge engine
- API client: `pip install wolframalpha`



Israel



Demographics:

Show rates

Show dis

population	8.32 million people (world rank: 100 th) (2017 estimate)
population density	385 people/km ² (people per square kilometer) (world rank: 34 th) (2017 estimate)
population growth	1.93 %/yr (world rank: 72 nd) (2017 estimate)
life expectancy	81.8 years (world rank: 21 st) (2013 estimate)
median age	30.2 years (world rank: 86 th) (2015 estimate)

Wikipedia

- Over 3,500 articles have been designated by the Wikipedia community as featured articles, exemplifying the best articles in the encyclopedia.
- Source : www.wikipedia.org
- Methods:
 1. Use it in the website
 2. Package: Wikipedia, MediaWikiAPI, WikipediaAPI

Wikipedia's sister projects

Wikipedia is hosted by the [Wikimedia Foundation](#), a non-profit organization that also hosts a range of other [projects](#):



Commons

Free media repository



Wikibooks

Free textbooks and manuals



Wikiquote

Collection of quotations



Wikiversity

Free learning materials and activities



MediaWiki

Wiki software development



Wikidata

Free knowledge base



Wikisource

Free-content library



Wikivoyage

Free travel guide



Meta-Wiki

Wikimedia project coordination



Wikinews

Free-content news



Wikispecies

Directory of species



Wiktionary

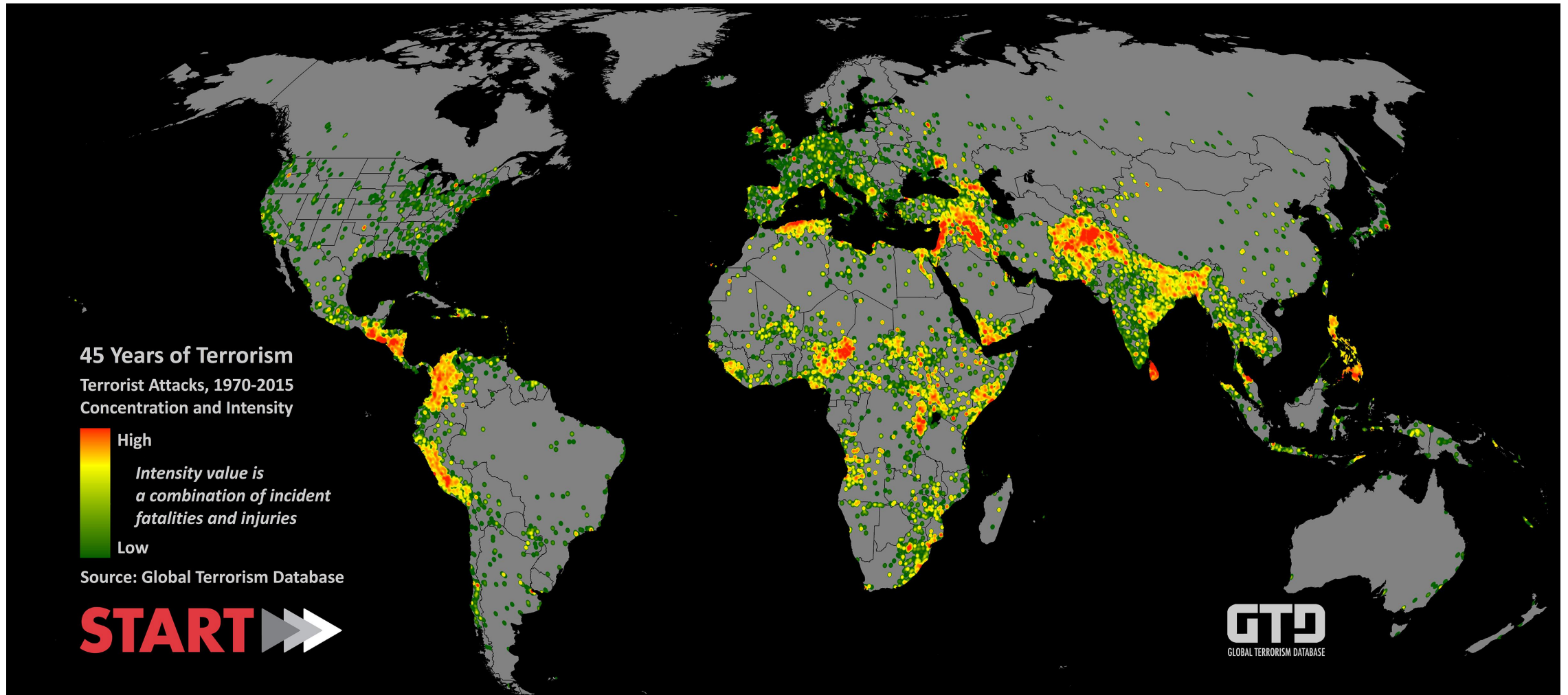
Dictionary and thesaurus

GTD

- The Global Terrorism Database (GTD) is an open-source database including information on domestic and international terrorist attacks around the world from 1970 through 2019.
- For each event, information is available on the date and location of the incident, the weapons used and nature of the target, the number of casualties, and—when identifiable—the group or individual responsible.

Description	Value
Records	>200,000
Time Range	1970~2019
Attributes	Timestamp, casualties, attack target, attack type, etc.
Source	https://www.start.umd.edu/gtd/

GTD



Geonames

- Source : <https://www.geonames.org>
- Contains over 25 million geographical names
- Including name, alternate names, location class, population, etc.
- Guides : Use it online and download it offline.

Zhangjiang - to view map click on map icon in bottom toolbar. (we need to reduce the cost for the map views) ×

Feature Hierarchy History Tags Alternate names

id 1785187

name Zhangjiang (preferably english or international name)

class city, village,... (P) ↕

code populated place (PPL) ↕

elevation m.

population

public tags

timezone Asia/Shanghai (CN) ↕

alternate countries (additional country codes for toponyms belonging to multiple countries, comma separated)

☒ ☐

News and Social Media

Chinese financial news

- 来源：东方财富和新浪财经；
- 时间：2020年5月-2020年12月；
- 数量：141423条；
- 数据链接：[Course Server](#)

- 数据格式：
 1. 新闻编码
 2. 新闻发布时间
 3. 新闻来源
 4. 新闻标题
 5. 新闻内容

	tid	datetime	source	title	text
0	'fb59985323056c7b84f5fe012148b6de	2020-05-03 05:34:00	新浪财经	巴菲特：欠福西博士一个巨大的感谢	沃伦-巴菲特（Warren Buffett）感谢白宫健康顾问安东尼-福西（Antho...
1	'cadedf264030d7248bcb7028c2eb7b370	2020-05-03 09:11:00	中国经济网	美法院坚持起诉刘强东，刘强东卸任多公司职务，只为不牵连京东！	说起刘强东的话，各位朋友们肯定都不会感到陌生吧，刘强东在国内乃至世界上都是著名的企业家，...
2	'b527d710f19f8f39abde295bcbbb8a3c	2020-05-03 07:27:00	新浪财经	外资控股券商扩容 同台竞技下差异化突围	外资控股券商扩容，同台竞技下差异化突围 随着今年4月1日起证券公司股比限制正式取消...
3	'1823a0bded4bf3f0d88c22273fce8b20	2020-05-03 05:22:00	新浪财经	巴菲特：没有什么可以阻止美国重新崛起	沃伦-巴菲特（Warren Buffett）周六表示，即使美国经济已经受到冠状病毒疫...
4	'd168ff3e2c096c0908ad8a9ac09c0f13	2020-05-03 07:23:00	新浪财经	用本土化策略突破“众口难调”的全球化桎梏	用本土化策略突破“众口难调”的全球化桎梏 30年、200亿的投入，换取一个自主品牌...

Reuters news

- Available in table Reuters in [Course Server](#), but the content needs to be crawled according to the URL

Description	Volume
Records	9,381,157
Time Range	2007/01/01-2018/11/01
Title	/
Date	/
URL	/

Date	title	URL
2018-04-10 10:43:00	BRIEF-Interma Trade Estimates FY 2017 Net Loss At	http://www.reuters.com/article/brief-interm
2018-04-10 10:43:00	Saudi Aramco signs deals with French companies w	http://www.reuters.com/article/us-france-sa
2018-04-10 10:43:00	China developer Evergrande to invest \$16 bln in hig	http://www.reuters.com/article/china-everg
2018-04-10 10:42:00	White House adviser says 'good progress' is being	http://www.reuters.com/article/trade-nafta-
2018-04-10 10:42:00	BRIEF-China Gem Unit To Further Grant A Facility In	http://www.reuters.com/article/brief-china-
2018-04-10 10:41:00	Spain may step up stake sales in Bankia privatizatio	http://www.reuters.com/article/us-bankia-p
2018-04-10 10:41:00	BRIEF-Soundwill Holdings Says Chairman Transferr	http://www.reuters.com/article/brief-sound
2018-04-10 10:39:00	BRIEF-Tandem Diabetes Care And Movi SpA Annou	http://www.reuters.com/article/brief-tander
2018-04-10 10:39:00	BRIEF-Pou Sheng International Says Application Ma	http://www.reuters.com/article/brief-pou-sl
2018-04-10 10:39:00	BRIEF-Farhad Fred Ebrahimi Reports A 5.09 Pct Stak	http://www.reuters.com/article/brief-farhad
2018-04-10 10:39:00	BRIEF-SoFi Names Michelle Gill Chief Financial Offic	http://www.reuters.com/article/brief-sofi-na
2018-04-10 10:39:00	BRIEF-Freddie Mac Prices \$1 Billion Multifamily K-D	http://www.reuters.com/article/brief-freddi
2018-04-10 10:38:00	BRIEF-Alantra Private Equity Completes Acquisition C	http://www.reuters.com/article/brief-alantra
2018-04-10 10:38:00	Spain may step up stake sales in Bankia privatisatio	http://www.reuters.com/article/bankia-priv
2018-04-10 10:38:00	BRIEF-Landing International Development Enters Le	http://www.reuters.com/article/brief-landin
2018-04-10 10:38:00	WHO says Australia exported listeria-tainted melons	http://www.reuters.com/article/us-australia
2018-04-10 10:37:00	White House adviser says Trump could seek coalitic	http://www.reuters.com/article/us-usa-trad
2018-04-10 10:36:00	WHO says Australia exported listeria-tainted melons	http://www.reuters.com/article/australia-list

NYT news

- More than 2,000,000 news
- Including URL, abstract, keywords, date etc.
- Available from the [NYT API](#)
- We have 2019/1 to 2020/2 New York Times data. 50,000+. [Course Server](#)

```
_id : 5bd50b214c0dbd457e11e4ac
web_url : https://www.nytimes.com/2000/01/01/world/yeltsin-resigns-the-economy-heartened-investors-give-stock-market-a-lift.html
snippet : President Boris N. Yeltsin's resignation today drove Russian stocks to their highest levels since the 1998 economic collapse, reflecting hope among investo...

lead_paragraph : President Boris N. Yeltsin's resignation today drove Russian stocks to their highest levels since the 1998 economic collapse, reflecting hope amon...

abstract : Russian stock prices rise to their highest levels since 1998 economic collapse in expectation that Vladimir V Putin, new acting president, will revive stalled ...

print_page : 19
> blog : (Array) 0 Elements
source : The New York Times
> multimedia : (Array) 0 Elements
> headline : (Document) 2 Fields
  main : Heartened Investors Give Stock Market a Lift
  kicker : YELTSIN RESIGNS: THE ECONOMY
  +
> keywords : (Array) 4 Elements
  > 0 : (Document) 2 Fields
    name : persons
    value : PUTIN, VLADIMIR V
    +
  > 1 : (Document) 2 Fields
  > 2 : (Document) 2 Fields
  > 3 : (Document) 2 Fields
  +
pub_date : 2000-01-01T00:00:00Z
document_type : article
news_desk : Foreign Desk
section_name : World
subsection_name : (Null)
> byline : (Document) 2 Fields
  type_of_material : News
  word_count : 294
  slideshow_credits : (Null)
  time : 2000-01-01 00:00:00.000
```

Jpost news

- Available in table Jpost in [Course Server](#)

Description	Volume
Records	341,478
Time Range	2010/01/01-2019/07/08
Title	/
Date	/
URL	/
News Body	/

title	Date	url	body
Sderot children send New Year's message to G	2010-01-01	www.jpost.com/Is	As 2009 drew to a close and hundreds of pro-Palestinian
Neturei Karta makes rare visit to Gaza	2010-01-01	www.jpost.com/N	A small group of anti-Zionist Jews were preparing Friday
Police expand search for missing Kiryat Tivon t	2010-01-01	www.jpost.com/B	Police expanded their search for a Kiryat Tivon teen on Fr
PA security forces arrest 5 settlers trying to tra	2010-01-01	www.jpost.com/B	Palestinian Authority security forces arrested five settlers t
Sanaa faces probe for assisting terror	2010-01-01	www.jpost.com/C	MK Taleb a-Sanaa (United Arab List-Ta'al) must stand tria
Fallout from gay debate rocks YU	2010-01-01	www.jpost.com/Je	A recent public forum at Yeshiva University that took on tl
UN to pull some international staff from Pakist	2010-01-01	www.jpost.com/B	The United Nations said it would relocate about a quarter
Report: Palestinian Authority denies Abbas rece	2010-01-01	www.jpost.com/B	Palestinian Authority Mahmoud Abbas has not received a
Ra'anana: Woman, 65, moderately hurt after dr	2010-01-01	www.jpost.com/B	A 65-year-old woman was moderately injured on Friday e
Aksa Martyrs Brigades claims responsibility for	2010-01-01	www.jpost.com/B	The Aksa Martyrs Brigades has claimed responsibility for
Nat'l infrastructures minister declares water pri	2010-01-01	www.jpost.com/B	National Infrastructures Minister Uzi Landau (Israel Beitei
Arabs, leftists demonstrate in Bil'in, Na'al	2010-01-01	www.jpost.com/B	Once again, dozens of Arabs and left-wing activists held v
Pakistan: Suicide bomb kills 88 at volleyball s	2010-01-01	www.jpost.com/B	A suicide bomber set off an explosives-laden vehicle on a
'Polish Jewish sites falling into ruin'	2010-01-01	www.jpost.com/Je	Hundreds of important Jewish historical and religious site
Hamas spokesman: No decision yet on Israeli p	2010-01-01	www.jpost.com/B	Hamas has yet to make a decision on Israel's latest propo

Twitter

- Donald Trump and 20 News agencies from more than 6 countries
- Available in table Twitter in [Course Server](#)

Description	Volume
Records	256,671
Time Range	2018~
ID	/
Content	/
Date	/
Source	/

_id	text	created_at	twitter_id
1067989026915799041	Revelers faced frigid temperatures at the #macysthank	2018-11-29T03:50:00.000Z	Reuters
1067988032358400000	RT @Breakingviews: Tiffany has stumbled in the luxury	2018-11-29T03:46:03.000Z	Reuters
1067986507682656256	VW banks on nostalgia and Jeep sells adventure at the	2018-11-29T03:40:00.000Z	Reuters
1067984850735587329	From @ReutersOpinion: Donald Trump flew off in a hu	2018-11-29T03:33:25.000Z	Reuters
1067984256922787840	RT @ReutersOpinion: Commentary: The best hope for	2018-11-29T03:31:03.000Z	Reuters
1067984154917306368	RT @ReutersTV: See today's editor's picks: https://t.co/	2018-11-29T03:30:39.000Z	Reuters
1067983991494586369	Ramen bar offers an oasis for New York's introverts ht	2018-11-29T03:30:00.000Z	Reuters
1067982387596935168	Elon Musk ducks out of one Los Angeles tunnel but pu	2018-11-29T03:23:38.000Z	Reuters
1067982092007604224	RT @Breakingviews: President Trump's deployment of	2018-11-29T03:22:27.000Z	Reuters
1067981474459086849	The stock market rallies after Fed Chairman Jerome Po	2018-11-29T03:20:00.000Z	Reuters
1067980036316323840	On @Breakingviews: Drugmaker Moderna's potential \$	2018-11-29T03:14:17.000Z	Reuters
1067978957822148608	LA-based URB-E says its folding electric scooter will br	2018-11-29T03:10:00.000Z	Reuters
1067976441948585984	The final Volkswagen Beetle model is one of the highli	2018-11-29T03:00:00.000Z	Reuters
1067976255608422400	Volkswagen planning new North America factory for el	2018-11-29T02:59:16.000Z	Reuters
1067971500605624322	Trump told Mueller he had no prior knowledge of 201	2018-11-29T02:40:22.000Z	Reuters

Financial Data

Crunchbase

- Crunchbase is a platform for finding business information about private and public companies.
- Crunchbase information includes investments and funding information, founding members and individuals in leadership positions, mergers and acquisitions, news, and industry trends.
- Crunchbase 2013 snapshot is available in [Course Server](#)

Variable	Volume
People	226709
Company	196553
Investment	80902
Degree Records	109610
Acquisition	9546
IPO	1259
Time Range	2007/01/06-2013/12/12

id	acquisition_id	acquiring_company_id	acquired_object_id	term_code	price_amount	price_currency	acquired_at	source_url	source_description	created_at	updated_at
1	1	c:11	c:10	(Null)	20000000	USD	2007-05-30	http://venturebeat.com/2007-05-30/Fox-Interactive-confirms-purchase-of-Googles-Android-OS/	2007-05-31 22:19:54	2008-05-21 19:23:44	
2	7	c:59	c:72	cash	60000000	USD	2007-07-01	http://www.techcrunch.com/2007-07-01/Deal-is-Confirmed-Google-Acquires-Android-OS/	2007-07-03 08:14:50	2011-05-06 21:51:05	
3	8	c:24	c:132	cash	2800000000	USD	2007-05-01	http://www.techcrunch.com/2007-05-01/CBS-Acquires-Europes-La-5-Realty-Group/	2007-07-12 04:19:24	2008-05-19 04:48:50	
4	9	c:59	c:155	cash	1000000000	USD	2007-06-01	http://techcrunch.com/2007-06-01/\$100-Million-Payday-For-Fox-Interactive-Confirms-Purchase-Of-Googles-Android-OS/	2007-07-13 09:52:59	2012-06-05 03:22:17	
5	10	c:212	c:215	cash	25000000	USD	2007-07-01	http://blog.seattlepi.nwso.com/2007-07-20/052907	2007-07-20 05:29:07	2008-02-25 00:23:47	
6	11	c:227	c:228	(Null)	7140000	USD	2003-08-05	http://www.businessweek.com/2003-08-05/E-Society-My-World-Is-Coming-From-Outside/	2007-07-20 09:58:29	2009-04-07 22:02:37	
7	12	c:5	c:230	cash	(Null)	USD	2007-07-01	http://www.techcrunch.com/2007-07-01/Deal-is-Confirmed-Google-Acquires-Android-OS/	2007-07-26 07:36:07	2009-08-10 19:37:37	
8	14	c:318	c:314	cash	(Null)	USD	2006-10-01	http://techcrunch.com/2006-10-01/Breaking-News-Cond-Nielsen-Buys-Rest-Of-NetR-2007-07-31-073836	2007-07-31 07:38:36	2012-01-16 03:04:04	
9	15	c:20	c:333	stock	1500000000	USD	2002-07-08	http://news.cnet.com/2002-07-08/eBay-picks-up-PayPal-for-2002-08-01-095826	2007-08-01 09:58:26	2008-10-06 19:04:29	
10	16	c:342	c:341	cash	3500000000	USD	2007-08-01	http://online.wsj.com/article/SB112342220070802091227	2007-08-02 09:12:27	2009-08-31 17:43:37	
11	17	c:24	c:347	cash	5000000	USD	2007-05-01	http://www.techcrunch.com/2007-05-01/CBS-Acquires-WallStrip-fo	2007-08-03 06:05:22	2008-05-19 04:48:50	
12	18	c:349	c:348	cash	130000000	USD	2007-03-01	http://techcrunch.com/2007-03-01/TechCrunch	2007-08-03 06:58:37	2011-06-10 21:34:22	
13	19	c:355	c:354	cash	3000000000	USD	2005-04-01	http://www.siliconindia.co/2005-04-01/	2007-08-03 13:53:49	2009-12-08 21:17:20	
14	20	c:377	c:100	cash	(Null)	USD	2005-05-29	http://www.there.com/pr_Makena-Technologies-Acquires-Android-OS/	2007-08-07 05:01:46	2011-08-22 00:03:07	
15	22	c:351	c:366	cash_and_stock	400000000	USD	2006-08-14	http://www.techcrunch.com/2006-08-14/	2007-08-08 05:53:52	2009-06-29 12:21:16	
16	23	c:307	c:306	cash	(Null)	USD	2007-08-07	http://venturebeat.com/2007-08-07/Forbes-buys-social-booker	2007-08-08 12:17:59	2008-07-15 02:35:25	
17	24	c:212	c:394	cash	41000000	USD	2007-08-01	(Null)	(Null)	2007-08-09 14:22:53	2008-02-25 00:23:47
18	26	c:499	c:495	cash	25000	USD	2007-07-01	(Null)	(Null)	2007-08-21 11:07:49	2008-02-25 00:23:47
19	27	c:29	c:528	cash_and_stock	10000000000	USD	2005-08-11	http://docs.yahoo.com/docs/2005-08-11/YAHOO!-AND-ALIBABA-CONFIRM-2007-08-22-083326	2007-08-22 08:33:26	2009-05-20 05:18:35	
20	28	c:29	c:530	cash	160000000	USD	2004-09-01	(Null)	(Null)	2007-08-22 08:49:57	2008-02-25 00:23:48
21	29	c:541	c:52	cash	327000000	USD	2007-02-01	http://login.vnuemedia.com/2007-02-01/Nielsen-buys-rest-of-NetR-2007-08-22-105920	2007-08-22 10:59:20	2008-05-25 18:20:16	

Listed company fundamental data (新浪财经)

- 数据来源: <https://finance.sina.com.cn>
- 根据股票代码搜索, 根据自己需要爬取



A股	港股	美股
公司基本信息	公司基本信息	公司基本信息
十大股东	重要财务指标	持股机构
财务摘要	资产负债表	
财务指标	现金流量表	
资产负债表	综合损益表	
利润表	主要持有人权益变动	
现金流量表		
所有者权益变动表		
公司高管		

Analysts' reports (券商分析师研究报告)

- 券商宏观研报14365份
- 慧博 (www.hibor.com.cn)

行业研报包含:

- 电子信息
- 电子元件
- 化工行业
- 机械行业
- 食品饮料
- 输配电气
- 通讯行业
- 文化传媒
- 医药制造

数据链接:

宏观研报: [百度网盘](#)

2018年6月3日

通信行业



电信启动天通手机集采，持续看好卫星应用产业链

——通信行业 2018 年第 23 周周报

行业周报

◆中国电信计划采购 10 万台天通手机，加速卫星通信行业商用

中国电信发布天通手机集中采购项目招标公告，计划采购天通手机 10 万台，其中单模和双模产品各 5 万台。继 5 月 16 日“天通一号卫星移动通信”面向商用市场正式放号后，中国电信快速启动 10 万部天通手机的集采，表明在航天科技集团 Ka 高通量卫星投入使用情况下，电信在卫星移动通信业务的推进明显加速。另外，我国于 2016 年发射的“天通一号”卫星已经经过近 2 年的测试和地面配套的搭建，基础设施建设逐步成熟，卫星通信的规模化商用为相关卫星通信终端及配套厂商带来巨大的发展机遇，预计市场规模超 200 亿元。随着我国天通一号后续 02/03 卫星的发射升空，天通卫星覆盖范围将持续增大。

◆自主可控仍是主线，继续看好卫星应用产业

近期受市场系统性调整影响，卫星应用板块出现了不同程度的回调。我们认为，中美贸易摩擦、中兴通讯事件将更加彰显出必须大力发展科技才能强国，才能实现“进口替代”和“自主可控”的发展，而卫星应用

买入（维持）

分析师

田明华 (执业证书编号：S0930516050002)
0755-82541645
tianminghua@ebcn.com

联系人

石崎良
021-22169327
shiqi@ebcn.com

吴希凯

wuxk@ebcn.com

行业主要数据图

Listed company announcements

- 数据链接: [Course Server](#)
- 数量: 21871条
- 时间: 2018年
- 数据包含信息:
 1. 数据来源URL
 2. 交易所公告内容 (Sentences)
 3. 从内容中抽出的实体 (valid_mspans)
 4. 实体的类别 (ann_mspan2guess_field)
 5. 事件列表 (recguid_eventname_eventdict_list)

```
▼ 0: [] 2 items
  0: "SH603009_2018-08-04_1205257201"
  ▼ 1:
    ▼ sentences: [] 20 items
      0: "证券代码: 603009证券简称: 北特科技公告编号: 2018-066"
      1: "上海北特科技股份有限公司关于控股股东股份补充质押的公告"
      2: "本公司董事会及全体董事保证本公告内容不存在任何虚假记载、误导性陈述"
      3: "或者重大遗漏,并对其内容的真实性、准确性和完整性承担个别及连带责任。"
      4: "上海北特科技股份有限公司(以下简称“公司”)今日收到公司第一大股东靳坤先生的通知,靳坤先生将其持有的公司部分股份"
      5: "一、控股股东本次补充质押的具体情况"
      6: "靳坤先生将其持有的3500000股流通股股票补充质押给海通证券股份有限公司,补充质押期限为2018年8月2日至2019年5月2日"
      7: "本次补充质押股份数占公司总股本的0.97%。"
      8: "截至本公告披露日,靳坤先生及其一致行动人靳晓堂先生分别持有公司股票140865400股和27748755股,合计168614155股,押公司股票数量为115471250股,占公司总股本的32.15%。"
      9: "靳坤先生及其一致行动人靳晓堂先生累计质押公司股票数量为121471250股,占公司总股本的33.83%。"
      10: "二、股票质押目的、资金偿还能力及相关安排"
      11: "1、股票质押目的"
      12: "靳坤先生本次股票质押式回购交易目的为对此前股票质押式回购交易的补充质押,不涉及新增融资安排。"
      13: "2、资金偿还安排"
      14: "靳坤先生为公司控股股东,其个人资信状况良好,具备相应的资金偿还能力,未来还款资金来源主要包括股票分红、个人收"
      15: "三、可能引发的风险及应对措施"
      16: "本次股票质押式回购协议项下履约保障比例预警线和履约保障比例最低线分别为170%和150%,若因公司股价下跌,达到履约控股股东的上述股票质押被平仓或被强制平仓的可能性很小,后续如出现平仓风险,控股股东将采取追加保证金或提前还款等措"
      17: "特此公告。"
      18: "上海北特科技股份有限公司董事会"
      19: "二〇一八年八月六日"
    ► ann_valid_mspans: [] 13 items
    ► ann_valid_dranges: [] 23 items
    ► ann_mspan2dranges:
    ▼ ann_mspan2guess_field:
      603009: "StockCode"
      北特科技: "StockAbbr"
      上海北特科技股份有限公司: "CompanyName"
```

Market data

- Source:

- Go to <https://finance.yahoo.com>
- Enter what you want to search into the search field
- Click Historical Data;
- Click Apply; Click Download Data.

- Other Download Methods:

- pip install yahoo-finance
- pip install ystockquote

- Attributes: Date, open, high, low, close, volume (daily)

The screenshot shows the Yahoo Finance website interface. At the top, there's a navigation bar with links like Home, Mail, News, Finance, Sports, Entertainment, Search, Mobile, and More... Below this is the Yahoo Finance logo and a search bar. The main content area displays various market indices (S&P Futures, Dow Futures, Nasdaq Futures, Russell 2000 Futures, Crude Oil) and a section for Target Corporation (TGT). The TGT section shows the current price at 186.09, a change of +2.65 (+1.44%), and a pre-market price of 187.50 (+1.41 (0.76%)). Below this, there's a tabbed interface with 'Historical Data' selected. The 'Historical Data' section shows a time period of Mar 02, 2020 - Mar 02, 2021, with 'Show: Historical Prices' and 'Frequency: Daily'. A red box highlights the 'Apply' button and the 'Download' link. Below the controls is a table of historical data.

Date	Open	High	Low	Close*	Adj Close**	Volume
Mar 01, 2021	185.95	187.58	184.85	186.09	186.09	3,655,400

天眼查数据

- 30家中国外汇交易中心人民币外汇即期做市商及其投资公司
- 数量：700+公司
- 包含信息：
 - 风险信息包含包括人员周边/预警风险信息
 - 对外投资包括被投资企业、企业法人、投资占比、对外投资总数等字段的详细信息
 - 工商信息包括公司名称或ID、类型、成立日期、经营状态、注册资本、法人、工商注册号、统一社会信用代码、组织机构代码、纳税人识别号等字段信息
- 链接：[Course Server](#)

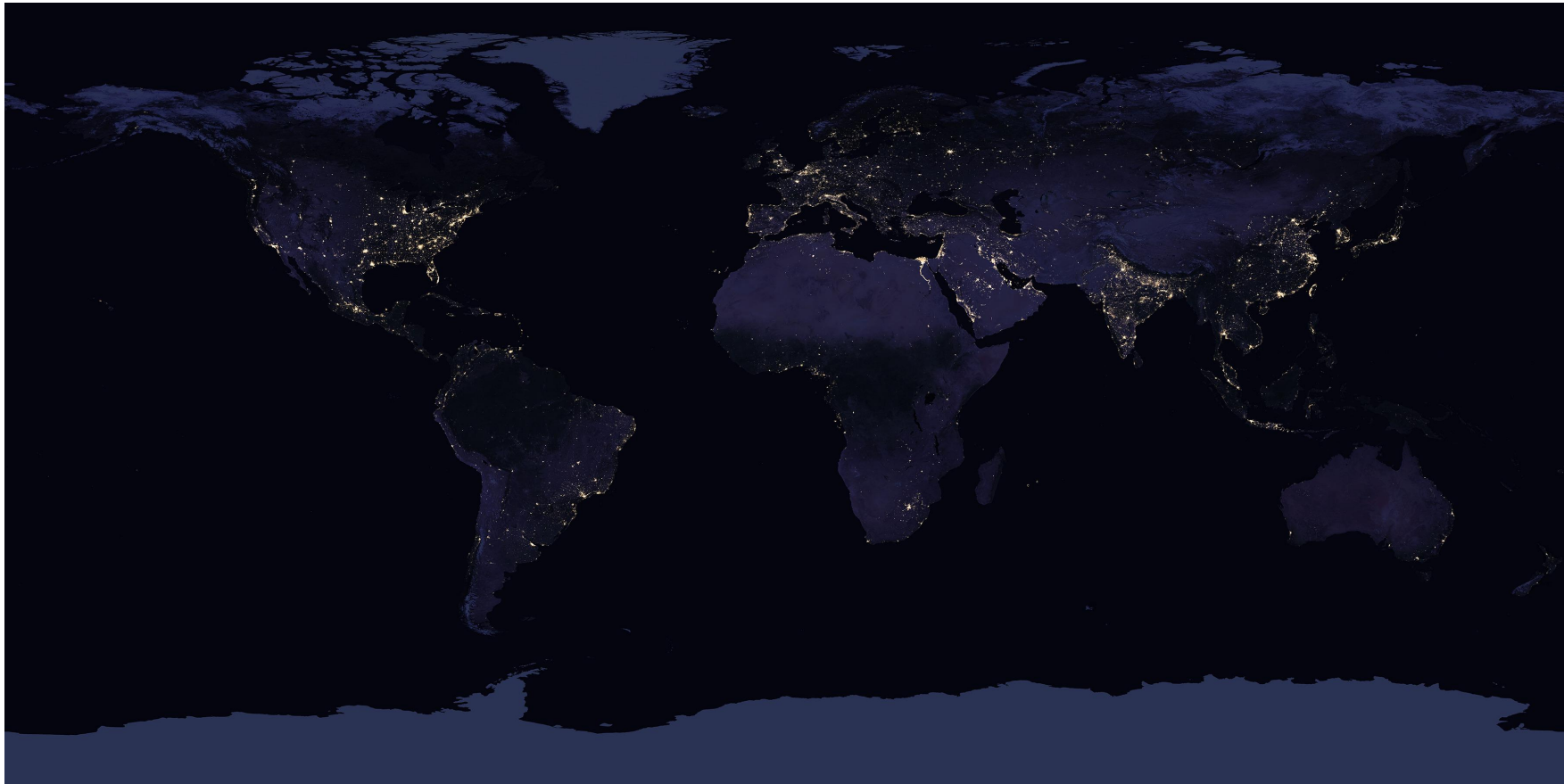
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  ],
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    "山东雅都建筑装饰有限公司",
    3,
    "的股东"
  ],
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    "日照东辰企业集团有限公司",
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            "title": "被执行人",
            "tag": "高风险信息",
            "list": [
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                "title": "该公司 <em>曾因未按时履行法律义务而被法院强制执行</em>",
                "desc": "被执行人",
                "type": 5,
                "riskCount": 1
              }
            ]
          }
        ]
      }
    ],
  },
},
```

Socio-Economic Data

Nighttime light intensity

- Source: <https://earthobservatory.nasa.gov/features/NightLights>
- Introduction: The data was collected using polar orbiting satellites that provide full cover of the globe twice a day



Socioeconomics data

- Source:
 - <https://www.earthdata.nasa.gov/data/catalog>
- The Socioeconomic Data and Applications Center provides data sets of human interactions in the environment, like
 - Global Subnational Infant Mortality Rates, v2.01 (2015)

The Global Subnational Infant Mortality Rates, Version 2.01 consist of Infant Mortality Rate (IMR) estimates for 234 countries and territories, 143 of which include subnational units.
 - Basic Demographic Characteristics, v4.11 (2010)

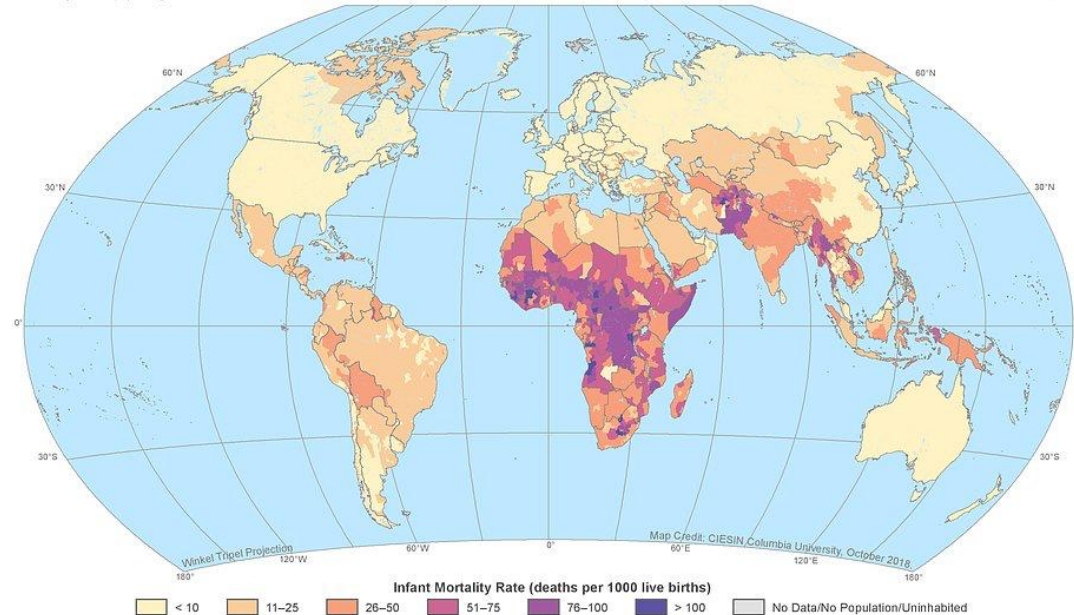
To provide estimates of population counts by age and sex for the year 2010, consistent with national censuses and population registers, as raster data to facilitate data integration.

Socioeconomics data

Global Subnational Infant Mortality Rates, Version 2

Poverty Mapping

2015



Global Subnational Infant Mortality Rates, Version 2 is part of the Poverty Mapping collection. This map displays infant mortality rate (IMR) estimates for 234 countries and territories, 143 of which include subnational units, at a spatial resolution of 30 arc-seconds (~1 km) for the year 2015.

Center for International Earth
Science Information Network
EARTH INSTITUTE | COLUMBIA UNIVERSITY

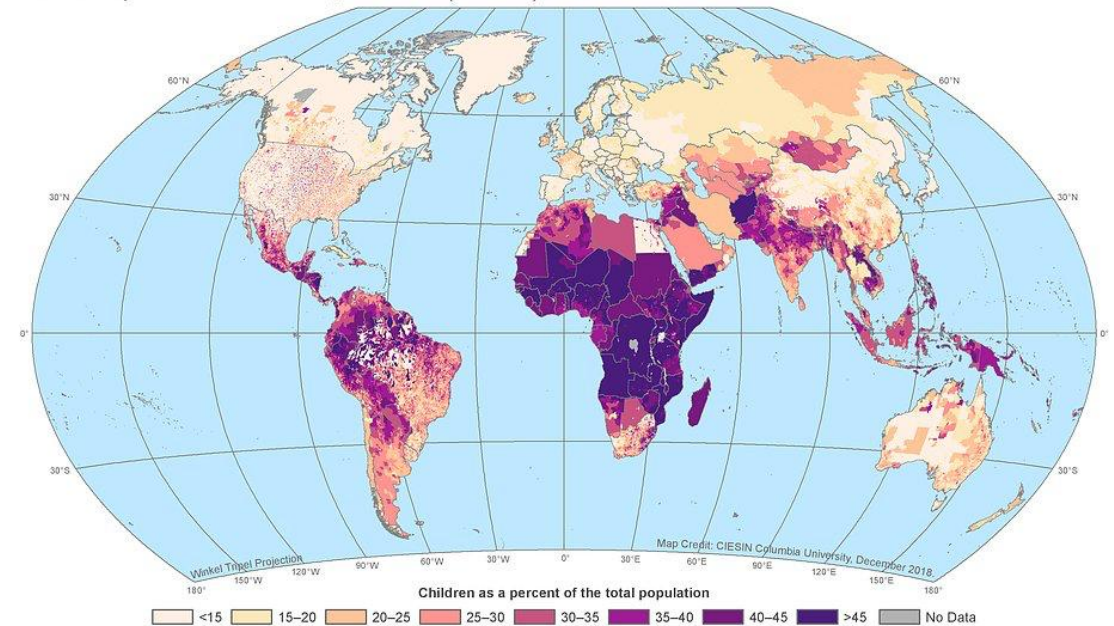
Data Source: Center for International Earth Science Information Network - CIESIN - Columbia University, 2018. Global Subnational Infant Mortality Rates, Version 2. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC).
<https://doi.org/10.7927/H4PN93JJ>.

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Basic Demographic Characteristics, v4.11, 2010: Children (Ages 0–14)

Gridded Population of the World, Version 4 (GPWv4)



Gridded Population of the World, Version 4 (GPWv4) Basic Demographic Characteristics, Revision 11 consists of estimates of human population by age and sex consistent with national censuses and population registers for the year 2010. The global distribution of children, ages 0–14, is represented here as a percent of the total population in the year 2010. It is calculated by summing the counts in the v4.11 5-year age group rasters from ages 0 to 14, then dividing the sum by the total population in the year 2010 and multiplying the quotient by 100.

Center for International Earth Science Information Network - CIESIN - Columbia University, 2018. Gridded Population of the World, Version 4 (GPWv4): Basic Demographic Characteristics, Revision 11. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC). <https://doi.org/10.7927/H46M34XX>.

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